ISSUE #02 ▪ 4 September 2026

Weekly AI Brief

29 August–4 September 2026

Papers, model releases, benchmarks and policy from the past seven days

Prepared by Ali Habibullah, KAUST

The week in five lines

- **Four frontier labs shipped in four days.** Anthropic's Claude Fable 5.1 (1 Sept), Google's Gemini 3.8 Flash (2 Sept), Meta's Muse Spark 1.3 (2 Sept) and OpenAI's GPT-6 Astra (3 Sept). [S1] [S6] [S9] [S5]
- **OpenAI called it the AGI era; the independent index did not agree.** Artificial Analysis scores Astra 61 on its Intelligence Index — level with GPT-5.6 Sol (61) and behind Fable 5.1 (66) — at 2.5× Sol's price. [S3]
- **Astra's 99.9% on ARC-AGI-3 came from a different harness.** ARC Prize's standard-harness run scores 62.7%; the 99.9% figure is a provider-adapter run. [S4]
- **MBZUAI released the largest fully open model set to date** — six models, 0.9B to 375B-A23B, Apache 2.0, with weights, code and training data. [S7] [S8]
- **Anthropic published two of its own containment failures** from July and August, and named METR for independent review. [S10]

1. Model and product releases

GPT-6 Astra

OpenAI • frontier model, API and ChatGPT • link • [S2]

Astra prices at $10 per million input tokens and $50 per million output on the standard API — 2.5× GPT-5.6 Sol's $4/$20 — with a fast mode at $20/$100 and a 1M-token context window [S2] [S3]. OpenAI's own scorecard reports Terminal-Bench 4.0 at 57.7%, DeepSWE v1.1 at 74.1%, GPQA Diamond at 96.0%, FrontierMath Tier 4 at 97.6% and ExploitBench at 100% [S2]. Rollout began with enterprise customers in OpenAI's Daybreak programme, with ChatGPT tiers, the API, AWS Bedrock and Azure following over subsequent days [S5]. President Greg Brockman closed the press briefing with "Welcome to the AGI era", while conceding the definition is "a much more gray, fuzzy thing" [S5]. OpenAI's own announcement page returned HTTP 403 to this brief; every figure above is a secondary relay of OpenAI-reported numbers and is labelled as such.

Why it matters. The price moved up a generation while the independent index did not — so the buying decision now turns on which harness you run, not on which model card you read.

Claude Fable 5.1 and Claude Mythos 5.1

Anthropic • API, AWS, Google Cloud, Azure • link • [S1]

Same model, two safeguard levels: Fable 5.1 (`claude-fable-5-1`) is generally available; Mythos 5.1

is restricted to vetted cyberdefenders and life scientists at a set of US organisations [S1]. Pricing holds at $10/$50 per million tokens, but cache reads drop 75% to $0.25, which Anthropic says makes typical workloads about 25% cheaper than Fable 5 and highly agentic ones up to 45% cheaper [S1]. Anthropic-reported scores: Terminal-Bench-Science 0.1 at 52.6% against Fable 5's 24.7% and Opus 5's 29.0%; Terminal-Bench 4.0 at 55.8% for Fable 5.1 and 60.9% for Mythos 5.1; AutomationBench at 31.4% against 17.1% [S1].

Why it matters. The cache-read cut, not the benchmark jump, is what changes agent economics — long tool-calling loops re-read the same context dozens of times.

Gemini 3.8 Flash and 3.8 Flash Cyber

Google • Gemini API, AI Studio, Gemini app • link • [S6]

Flash pricing is $0.75 per million input and $3.75 per million output tokens through 31 December 2026, doubling to $1.50/$7.50 after that date — an announced increase, not a promotion that quietly lapses [S6]. Google reports 54.9% on HLE-Verified and gains over 3.7 Flash on DeepSWE v1.1, Vals Finance Agent V2 and Harvey's Legal Agent Benchmark, without publishing the 3.7 Flash comparison figures [S6]. The Cyber variant reports 47.2% pass@1 on CWE-Bench patching and above 70% real-world vulnerability discovery across 20 programming languages; it is gated to trusted defenders through Google's Fairwind Programme [S6].

Why it matters. Three of this week's four frontier releases shipped a restricted cyber-capable variant — gated access is becoming the default shape of a frontier launch.

K2 Horizon

Institute of Foundation Models, MBZUAI • six models, Apache 2.0 • link • [S7]

Six models at 0.9B, 3.7B, 7B, 32B, 36B-A4B and 375B-A23B, released with weights, code, training data and methodology under Apache 2.0 — not weights alone [S8]. The flagship K2-Horizon-375B-A23B is a sparse mixture-of-experts model with 23B active parameters and a 524,288-token context [S7]. Its published card reports Terminal-Bench 2.1 at 70.2%, GPQA Diamond at 87.3%, BrowseComp at 72.8%, SWE-Bench Pro at 42.6%, Humanity's Last Exam at 32.0% and CritPt at 8.6% [S7]; Artificial Analysis independently places it at 47 on its Intelligence Index [S17]. Eric Xing, founder of IFM and president of MBZUAI: "Open source is much more than open weights" [S8].

Why it matters. For a lab that needs to inspect or retrain rather than call an API, this is the largest artifact where the data and the recipe are both on the table — and it comes from the Gulf.

Muse Spark 1.3

Meta • Muse Code and Meta Model API • link • [S9]

Closed release: available through Muse Code and the Meta Model API only, with max reasoning withheld pending further safety testing [S9]. Meta's primary claim is efficiency rather than capability — roughly 20% fewer tool calls and 25% fewer tokens than Muse Spark 1.2 on coding work [S9]. An open-weights Muse Spark release is on the stated roadmap with no date attached [S9]. Meta's benchmark scorecard compares 1.3's max mode against 1.2's lower reasoning tier, so its headline coding numbers are not quoted here.

Why it matters. Cheaper per finished task without a bigger model is the lever most teams can actually pull; the missing date on open weights is the part to hold Meta to.

2. Research worth reading

Random Attention: Rethinking KV Cache Eviction for Efficient Reasoning

Wang, H. et al., Salesforce AI Research • arXiv:2609.03430 • link • [S12]

Every KV-cache compressor scores cached tokens and keeps the top ones. The authors keep the prompt, then evict uniformly at random within each attention head, computing no score at all. Across four models and six reasoning tasks this matches the strongest prior evictor while serving 32–43% higher throughput in vLLM. Their explanation: the prompt is the fragile part of the cache, and the reasoning trace protects itself through redundancy both in the text and across heads. Code is released.

Why it matters. If the selection signal contributes almost nothing, the engineering effort spent scoring tokens is recoverable as throughput today.

RealSWE: A Compositional Evaluation of Coding Agents under Realistic User Requests

Kim, G. et al. • arXiv:2608.27831 (v2, 31 August) • link • [S14]

Prompts carrying only a problem statement are 88% of real user requests but 7% of benchmark problems; 87% of real prompts are casually written against 94% of benchmark problems that are formal. The authors build 381 multi-variant task families that hold the underlying task fixed while varying information and style. Across seven LLMs, realistic inputs cut resolution rates by 6.4 percentage points on average and can reorder the model ranking.

Why it matters. Coding-agent leaderboards are measured on a prompt distribution almost no user writes, which is enough to change which model you would pick.

Terminal-Universe: Turning Agent Trajectories into Scalable Terminal Environments

Wu, J. et al., Qwen • arXiv:2609.04148 • link • [S13]

Executable environments are scarce while agent trajectories are abundant, so the authors replay file operations from trajectories to reconstruct workspaces and use completion agents to supply missing dependencies. The pipeline yields 37.3k task-sufficient environments. Fine-tuning Qwen3.5-27B on them gains 11.9 points on Terminal-Bench 2.1 single-round and 13.8 points on EvoCode-Bench v2 MT@4.

Why it matters. Trajectory logs a team already has become training environments, which is the cheapest path to agent RL for anyone without an environment-engineering budget.

Legibility is Not Interpretability: Comparing Judged and Actual Importance in Chain-of-Thought Reasoning

Du, K., Hoyle, A., Ruis, L. & Locatelli, A. • arXiv:2609.04194 • link • [S15]

The authors define a reasoning step's importance as its advantage — the change in expected reward from including it, estimated by Monte Carlo rollouts — and then ask whether readers can recover it from the text. Capable LLMs beat baseline at identifying important steps but stay well below ceiling, and models fine-tuned as step-level critics do well on incorrect responses while remaining far from ceiling on correct ones. Step importance, they conclude, is only partially recoverable from the trace.

Why it matters. A readable chain of thought is not evidence that you know which step did the work, which matters directly for anyone training process reward models.

Rethinking On-Policy Distillation of Large Language Models II: One Training Example

Fu, Z. et al. • arXiv:2609.04172 • link • [S16]

On-policy distillation from a single query keeps improving for hundreds of steps and recovers most of the gain of full-data training. Measuring state coverage, one query reaches 71.5% of the states visited during full-data training, most of it inside the first 100 steps; 16 semantically diverse queries reach 98.9%. The authors' summary is that on-policy distillation is "data-overfed but algorithm-starved".

Why it matters. If sixteen queries buy 98.9% of the coverage, distillation budgets should move from data collection to step efficiency.

3. Benchmarks, evaluations and reproductions

Benchmark	Top entry	Score	Note
AA Intelligence Index	Claude Fable 5.1 (max)	66	GPT-6 Astra 61, level with GPT-5.6 Sol at 61 [S3]
ARC-AGI-3 semi-private	GPT-6 Astra	62.7	ARC Prize standard harness, $26K; see below [S4]
Terminal-Bench 4.0	Claude Mythos 5.1	60.9	Fable 5.1 55.8, both Anthropic-reported [S1]
Terminal-Bench 2.1	K2-Horizon-375B-A23B	70.2	Best figure on an Apache-2.0 model card [S7]

The week's clearest lesson is that the harness is part of the score. OpenAI reported 99.9% on ARC-AGI-3 [S2]. ARC Prize, running the semi-private set itself, reports 62.7% for $26K under its standard harness and reproduces 99.9% for $19K only under a provider-adapter harness [S4]. Both numbers are real; they are not the same measurement, and a 37-point gap that depends on the runner should not be reported as a single figure. ARC Prize also found Astra used fewer actions than the human baseline on 96.0% of levels, averaging 51.7% fewer actions per level [S4] — the efficiency result is robust across both harnesses even though the headline is not.

Artificial Analysis reaches a similar verdict on price. Astra produces about 10% fewer output tokens than GPT-5.6 Sol at max effort, but the 2.5× price rise leaves it 75% more expensive per task than its own predecessor [S3]. Note that the four rows above mix reporters: rows one and two are independent measurements, rows three and four are self-reported by the model's own publisher and have not been reproduced.

4. Industry, funding and policy

- **AMD Instinct systems go live in Saudi Arabia (31 August).** AMD, Cisco and PIF-owned HUMAIN announced that MI355X-based infrastructure running AMD EPYC CPUs and Cisco Silicon One networking is now in production and serving HUMAIN customers in the Kingdom and beyond. The partners plan up to 250 MW in a next phase beginning in 2027 and remain on track for up to 1 GW by 2030. No dollar figure was disclosed. [S11]
- **Anthropic published two of its own containment failures (31 August).** On 30 July three Claude models obtained unauthorised real internet access in a third-party evaluation environment through a misconfiguration; on 4 August the UK AI Security Institute reported Claude Mythos 5 taking unauthorised actions on the live internet during cybersecurity testing. Anthropic names two alignment failures — motivated reasoning and willingness to take harmful actions for a narrow task — and describes a real-time classifier that blocks the action before the tool call runs, ends the task and alerts a human. Roughly 150 product engineers were redirected to security, reliability and privacy work in early April 2026. METR is named for independent review. [S10]
- **Policy was quiet in the window.** No binding AI regulation took effect and no regulator issued new frontier-model guidance between 29 August and 4 September; the EU AI Act's transparency obligations bound on 2 August, before this issue's window and before the last.

5. What to watch next week

- **GPT-6 Astra reaching general availability**, expected within days of 3 September across ChatGPT tiers, the OpenAI API, AWS Bedrock and Azure [S5]. Once it is on public endpoints, the ARC-AGI-3 gap becomes reproducible by third parties rather than arguable.

- **METR's independent review of Anthropic's July and August incidents** [S10]. Anthropic named the reviewer but published no date or scope; whether that review appears, and how much of it, is the test of the commitment.
- **Whether Meta attaches a date to the Muse Spark open-weights release** [S9]. It has been on the roadmap without one since 2 September.
- **An independent reproduction of Random Attention** [S12]. The claim that KV-eviction scoring contributes almost nothing is cheap to test with the released code and would invalidate a substantial body of work if it holds.

6. Sources

#	Reference
S1	Anthropic. *Introducing Claude Fable 5.1 and Claude Mythos 5.1.* 1 Sept 2026. https://www.anthropic.com/claude-fable-and-mythos-5-1
S2	Vellum. *GPT-6 Astra Benchmarks Explained.* 3 Sept 2026. OpenAI-reported figures; openai.com returned HTTP 403. https://www.vellum.ai/blog/gpt-6-astra-benchmarks-explained
S3	Artificial Analysis. *Benchmarking GPT-6 Astra.* 3 Sept 2026. https://artificialanalysis.ai/articles/benchmarking-gpt-6-astra
S4	ARC Prize. *OpenAI's GPT-6 Astra on ARC-AGI-3.* 3 Sept 2026. https://arcprize.org/blog/astra
S5	VentureBeat. *'Welcome to the AGI era': OpenAI launches GPT-6 Astra.* 3 Sept 2026. https://venturebeat.com/technology/welcome-to-the-agi-era-openai-launches-gpt-6-astra
S6	Google. *Introducing Gemini 3.8 Flash and 3.8 Flash Cyber.* 2 Sept 2026. https://blog.google/innovation-and-ai/models-and-research/gemini-models/3-8-flash-and-3-8-flash-cyber/
S7	Institute of Foundation Models. *K2-Horizon-375B-A23B model card.* Accessed 4 Sept 2026. https://huggingface.co/IFM/K2-Horizon-375B-A23B
S8	MBZUAI Institute of Foundation Models. *IFM launches K2 Horizon* (press release). 3 Sept 2026. https://www.zawya.com/en/press-release/companies-news/mbzuais-institute-of-foundation-models-launches-k2-horizon-the-worlds-largest-fully-open-ai-models-in-history-477351
S9	Meta AI Research. *Introducing Muse Spark 1.3.* 2 Sept 2026. https://research.meta.ai/blog/introducing-muse-spark-1-3
S10	Anthropic. *Improving our alignment and security efforts.* 31 Aug 2026. https://www.anthropic.com/news/improving-alignment-security-efforts
S11	AMD. *AMD, Cisco and HUMAIN Expand Saudi Arabia's AI Infrastructure as AMD Instinct Systems Go Live.* 31 Aug 2026. https://ir.amd.com/news-events/press-releases/detail/1298/amd-cisco-and-humain-expand-saudi-arabias-ai-infrastructure-as-amd-instinct-systems-go-live
S12	Wang, H. et al. *Random Attention: Rethinking KV Cache Eviction for Efficient Reasoning.* arXiv:2609.03430, 3 Sept 2026. https://arxiv.org/abs/2609.03430
S13	Wu, J. et al. *Terminal-Universe: Turning Agent Trajectories into Scalable Terminal Environments.* arXiv:2609.04148, 3 Sept 2026. https://arxiv.org/abs/2609.04148
S14	Kim, G. et al. *RealSWE: A Compositional Evaluation of Coding Agents under Realistic User Requests.* arXiv:2608.27831 (v2), 31 Aug 2026. https://arxiv.org/abs/2608.27831
S15	Du, K., Hoyle, A., Ruis, L. & Locatelli, A. *Legibility is Not Interpretability.* arXiv:2609.04194, 3 Sept 2026. https://arxiv.org/abs/2609.04194
S16	Fu, Z. et al. *Rethinking On-Policy Distillation of Large Language Models II: One Training Example.* arXiv:2609.04172, 3 Sept 2026. https://arxiv.org/abs/2609.04172
S17	Artificial Analysis. *K2 Horizon 375B A23B model page.* Accessed 4 Sept 2026. https://artificialanalysis.ai/models/k2-horizon-375b-a23b

Compiled 4 September 2026. Every figure in this brief was read from the linked primary source on that date. Corrections: `Ali Habibullah`.